

This assignment will not be graded and is only for practice.

Level 1:

1) Consider a square matrix $A = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & -1 \\ -1 & 0 & 1 \end{bmatrix}$. If **1 point**

$$P = -1M_{11} - 2M_{12} + 0M_{13}$$

$$Q = -2M_{21} + 0M_{22} + 1M_{23}$$

$$R = -1M_{31} - 0M_{32} + 1M_{33}$$

where M_{ij} is the minor with respect to the (i, j) -th entry, then choose the set of correct options.

☐ $\det(A) = -2$.

☐ $P = Q$.

☐ $P = Q = R$.

☐ $P = Q = R$.

☐ $Q = R$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\det(A) = -2.$$

$$P = Q = R.$$

2) Suppose for a real 3×3 matrix A , there exists a real 3×3 matrix P such that $D = PAP^{-1}$ is **1 point** a real 3×3 diagonal matrix. Choose the correct set of options.

☐ $\det(A)$ must be equal to $\det(P)$.

☐ $\det(A)$ must be equal to $\det(D)$.

☐ The matrix A must be equal to D .

☐ If D is the identity matrix of order 3, then A must be the identity matrix of order 3

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\det(A)$ must be equal to $\det(D)$.

If D is the identity matrix of order 3, then A must be the identity matrix of order 3

3) Choose the set of correct options for square matrices of order 3.

1 point

- ☐ If $AB = 0$, then one of the matrices between A and B must have determinant 0.
- ☐ If $AB = 0$, then both the matrices A and B must have determinant 0.
- ☐ If $AB = 3I$, where I denotes the identity matrix of order 3, then $\det A = \frac{3}{\det(B)}$.
- ☐ If $AB = 3I$, where I denotes the identity matrix of order 3, then $\det A = \frac{9}{\det(B)}$.
- ☐ If $AB = 3I$, where I denotes the identity matrix of order 3, then $\det A = \frac{27}{\det(B)}$.
- ☐ If $AB = 3I$, where I denotes the identity matrix of order 3, then both A and B must have non-zero determinant.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $AB = 0$, then one of the matrices between A and B must have determinant 0.

If $AB = 3I$, where I denotes the identity matrix of order 3, then $\det A = \frac{27}{\det(B)}$.

If $AB = 3I$, where I denotes the identity matrix of order 3, then both A and B must have non-zero determinant.

4) Consider a matrix $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$. If $a + b + c$ is divisible by 6 then choose the set of correct options.

1 point

- ☐ $\det(A)$ is divisible by 5.
- ☐ $\det(A)$ is divisible by 6.
- ☐ $\det(A)$ is divisible by 2.

☐ $\det(A)$ is divisible by 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\det(A)$ is divisible by 6.

$\det(A)$ is divisible by 2.

$\det(A)$ is divisible by 3.

5) Let $A = [a_{ij}]$ be a square matrix of order 3, where $a_{ij} = i$. Find $\det(A)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Level 2:

6) Let $A = [\delta_{ij}]$ be a square matrix of order 3, where $\delta_{ij} = i \times j$. Find $\det(A)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

7) Consider the following square matrices:

1 point

$$A = \begin{bmatrix} 2013 & 2014 & 2015 \\ 2016 & 2017 & 2022 \\ 2019 & 2020 & 2021 \end{bmatrix}, B = \begin{bmatrix} 2016 & 2017 & 2022 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix} \text{ and } C = \begin{bmatrix} 4032 & 4034 & 4044 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix}$$

Choose the set of correct options.

- ☐ $\det(B) = \det(A)$.
- ☐ $\det(A) = -\det(B)$.
- ☐ $\det(C) = -2\det(B)$.
- ☐ $\det(C) = -2\det(A)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\det(A) = -\det(B).$$

$$\det(C) = -2\det(B).$$

$$\det(C) = -2\det(A).$$

(Based on the below information answer questions 8 and 9).

Let A be a real 3×3 matrix as given below:

$$A = [C_1 \quad C_2 \quad C_3]$$

where C_i represents the i -th column of the matrix A . Now let us consider the following set of matrices.

$$A_1 = [C_1 \quad C_2 + 5C_3 \quad C_3]$$

$$A_2 = [C_1 + C_2 + C_3 \quad C_2 \quad C_3]$$

$$A_3 = [C_1 \quad C_2 + 5C_3 \quad 0]$$

$$A_4 = [C_1 + C_2 \quad C_2 + C_3 \quad C_3 + C_1]$$

8) Which of the matrices have the same determinant as that of A ?

1 point

- ☐ A_1
- ☐ A_2
- ☐ A_3
- ☐ A_4

No, the answer is incorrect.

Score: 0

Accepted Answers:

A_1

A_2

9) Choose the set of correct options.

1 point

- ☐ $\det(A_3)$ cannot be determined from the given information.
- ☐ $\det(A_3) = 0$
- ☐ $\det(A_4) = 2 \det(A)$
- ☐ $\det(A_2) = 3 \det(A)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\det(A_3) = 0$

$\det(A_4) = 2 \det(A)$

10) If the diagonal entries of a 3×3 lower triangular matrix A are 1,2, and 3, then find the sum of roots of the equation $\det(A - xI) = 0$, where I is the identity matrix of order 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

1 point

Your score is: 0/10